

Ben Amor Hanine

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Summary

Final-year ICT Engineering student at INSAT, experienced in classical ML, LLMs, RAG, computer vision and multimodal AI. Seeking a PFE internship in applied AI.

Professional Experience

AI & Data Science Intern (R&D) – Yonnov'IA

Marseille, France

August 2026 - September 2026

- Built a hybrid semantic recommendation engine (9 signals + MMR) via ONNX/fastembed + pgvector (HNSW/IVFFlat), over a full scraping-to-retrieval pipeline.
- Optimized the pipeline: latency $\sim 4-5s \rightarrow$ near-instant, memory footprint GB $\rightarrow \sim 100$ MB (PyTorch $\rightarrow$ ONNX Runtime), with PCA 3D explainability and precision@k evaluation.

AI Engineer (Part-Time) – 1MoreThing Ventures

Tunis, Tunisia

December 2025 – Present

ML & Computer Vision

June 2026 – Present

- Developed and validated 20+ CV use cases (detection, pose, tracking, OCR) with multi-step inference pipelines deployed on NVIDIA Jetson, under licensing and data privacy constraints.
- Fine-tuned object detection and classification models on custom annotated datasets, achieving 85% mAP@50.
- Deployed models on NVIDIA Jetson for edge inference.

AI, RAG, OCR

December 2025 – May 2026

- Designed an air-gapped, open-licence RAG system for a $\sim 1.25B$ -page Arabic judicial archive (Ministry of Justice); benchmarked OCR engines (PaddleOCR, Qari-OCR, Baseer) per page-type and quality tier via CER/WER on a handwriting-heavy, no-ground-truth corpus.
- Built an 8-stage pipeline (intake, audit, splitting, structuring, hybrid indexing) with hybrid retrieval (dense, BM25, visual) and cross-encoder reranking, using Qwen for answer generation.
- Enforced a strict no-answer-without-evidence rule via citation verification, graded abstention against hallucinations, and a verify-on-demand review loop with corrections persisted as training data.

Machine Learning Engineering Intern — HydatiS

Tunis, Tunisia

June 2025 – August 2025

- Architected an AI-powered personal safety platform integrating geospatial risk prediction and behavioral anomaly detection.
- Developed a voice AI pipeline using Whisper, TunBERT and SpeechBrain/Pyannote for transcription, classification and emotional analysis.
- Trained RandomForest/XGBoost models and deployed via FastAPI microservices with REST APIs for real-time alerts.

IT Instructor — PIZART Training Center

Sousse, Tunisia

June 2024 – Present

- Teaching Python, algorithms and ML for various levels; designing interactive educational projects.

Software Engineering Intern — ITGate Group

Sousse, Tunisia

July 2024

- Built a microservices-based RBAC system with Node.js, MongoDB, JWT and Angular for role management.

Technical Skills & Expertise

- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, AST, XGBoost, RandomForest
- **Generative AI & LLMs:** LLMs, RAG, Diffusion Models, Embeddings, Prompt Engineering, Qwen, LLaMA, Graded Abstention
- **Computer Vision:** Object Detection, Classification, Pose Estimation, Tracking, OCR, Edge AI, YOLO, RF-DETR, Fine-tuning, mAP Evaluation
- **Specializations:** NLP, Arabic NLP, Document AI, OCR (PaddleOCR), Video Processing, Speech & Audio AI (Whisper, SpeechBrain, TunBERT), Multimodal AI (ImageBind, BLIP-2)
- **Semantic Search & Vector ML:** pgvector, HNSW/IVFFlat, BM25, MMR, Cross-Encoder Reranking, Precision@k, Explainability (PCA)
- **MLOps & Deployment:** Docker, FastAPI, ONNX, Prometheus, Grafana, REST APIs, NVIDIA Jetson, MQTT
- **Data & Databases:** Pandas, NumPy, SQL, Spark, Kafka, HDFS, HBase, MapReduce, PostgreSQL, MySQL, MongoDB, Matplotlib/Seaborn, Power BI
- **Programming & Tools:** Python, Java, JavaScript/TypeScript, Git, GitHub, GitLab Actions, Linux, Agile

Projects

Intelligent Multimodal Approach for Automatic Generation of Short Formats (Shorts) *Publication in progress*
End-to-end highlight detection pipeline combining ImageBind, Whisper and BLIP-2 with a cross-modal weighting mechanism and Attention per segment. Zero-shot detection via transformer-based semantic reranking on generated descriptions, without labeled data.

Respiratory Sound Classification — AST & SAM Optimizer
Reproduced and improved a Transformer-based pipeline on ICBHI 2017; introduced Focal Loss, Mixup and patient-wise evaluation — ICBHI score 69.4%, outperforming the reference paper under stricter generalization conditions.

Big Data Pipeline — Chicago Air Quality Analysis
End-to-end pipeline: real-time ingestion via Kafka, Spark Structured Streaming to HDFS, batch analytics with Java MapReduce, serving via HBase and interactive Streamlit dashboard — deployed on Docker.

Optimization of Boltzmann Machines via Quantum Annealing *(contributor)*
Developed a QBM leveraging quantum annealing for training in large Hilbert spaces, with quantum state tomography on benchmark datasets.

Education

Engineering Degree in Computer Networks and Telecommunications 2027 | INSAT, Tunisia
Relevant Coursework: Machine Learning, Deep Learning, Data Analysis, Stochastic Processes, Optimization, Operations Research, Business Intelligence, Big Data

Preparatory Cycle 2024 | INSAT, Tunisia
High School Diploma in Mathematics (Highest Honors) 2022 | Sousse

Certifications

- AWS Certified AI Practitioner (AIF-C01) AWS | 2026
- Generative AI Foundations AWS | 2026
- ML Engineer Track Datacamp | 2025

Leadership & Extracurricular Activities

System Administrator — ACM INSAT 2023–2025

- Managed network infrastructure for 100+ participants across CodeQuest, WinterCup and DataQuest competitions.
- Mentored new members and co-organized the Data Overflow competition.

Languages

French | English | Arabic | Basic German and Spanish